

Introductory Concepts

Why Chemical Process Control

↳ necessity of process modeling

- Safety
- Environmental constraints
- Product Quality
- Energy Efficiency

Control Systems

↳ Maintain process conditions and desired values

by manipulating process variables to adjust variables of interest

- e.g. cruise control
- speed — controlled variable
 - desired value — set point
 - terrain, hills — disturbances
 - throttle, fuel flow — manipulated variables

home water heater

- water temp — controlled variable
- fuel flow to burner — manipulated variable
- heat losses, leaky faucets — disturbances

Disturbance Rejection $\Rightarrow$ ability to maintain controlled variables at set point in spite of disturbances

Set Point tracking $\Rightarrow$ ability to move process variables from one setting to new desired setting

$$\text{Error} = (\text{Set Point value}) - (\text{Measurement signal of Controlled variable})$$

Controller action $\rightarrow$ based on magnitude and sign of error $\rightarrow$ Provides input to process

Feed back Control → using information about deviation of the system from its desired state to control the system

Closed loop
feed back
(Automatic)

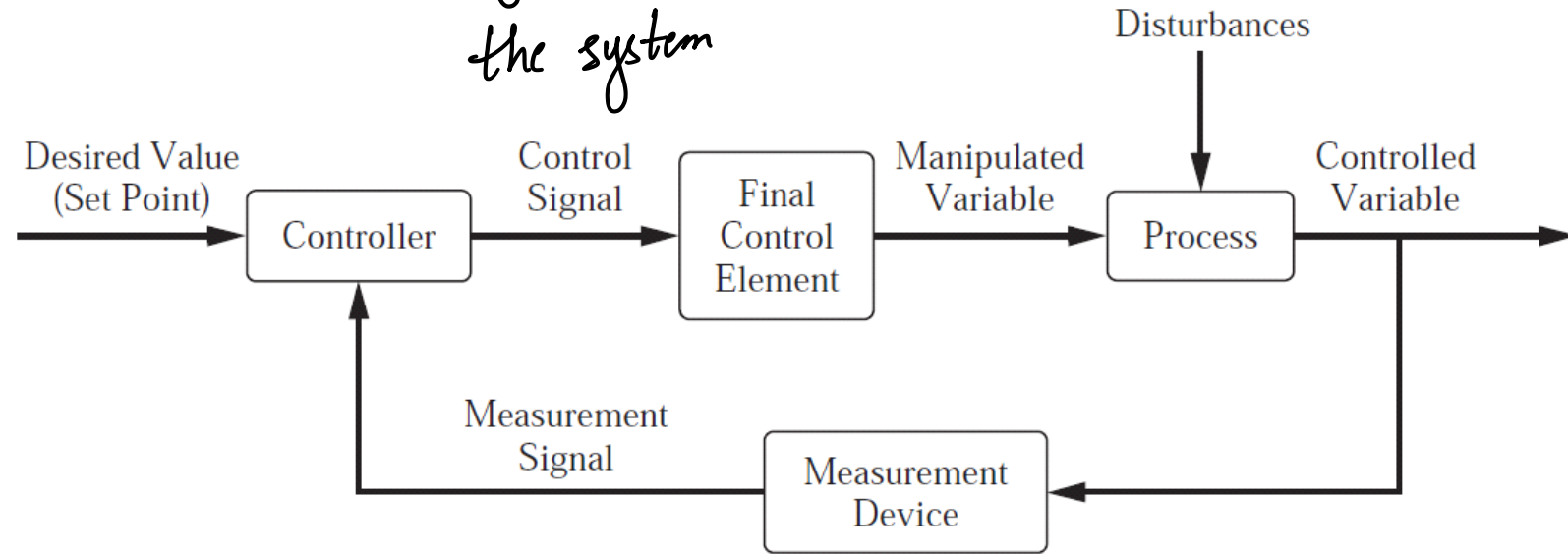


FIGURE 1-1
Generalized process control system.

Open loop system (Manual)

- ↳ Measurement signal disconnected from controller
- ↳ Controller output manually adjusted to change controlled variable

Negative Feedback

$$\text{Error signal} = \text{set point} - \text{measurement}$$

↳ the -ive measurement is added to (Fed back to) controller
setpoint to compute error

Example 1.1

Hot water tank controller

Set point — water temperature — $120-140^{\circ}\text{F}$

Measurement device — thermocouple — measures temp

Controller — Thermostat — computes error

$$\text{Error} = T_{\text{set point}} - T_{\text{measured}}$$

+++ Error $\Rightarrow$ temp lower than desired

$\hookrightarrow$ increase flow to burner

-ive Error $\Rightarrow$ temp higher than desired

$\hookrightarrow$ reduce flow to burner

Thermostat can increase or decrease fuel supply to burner.

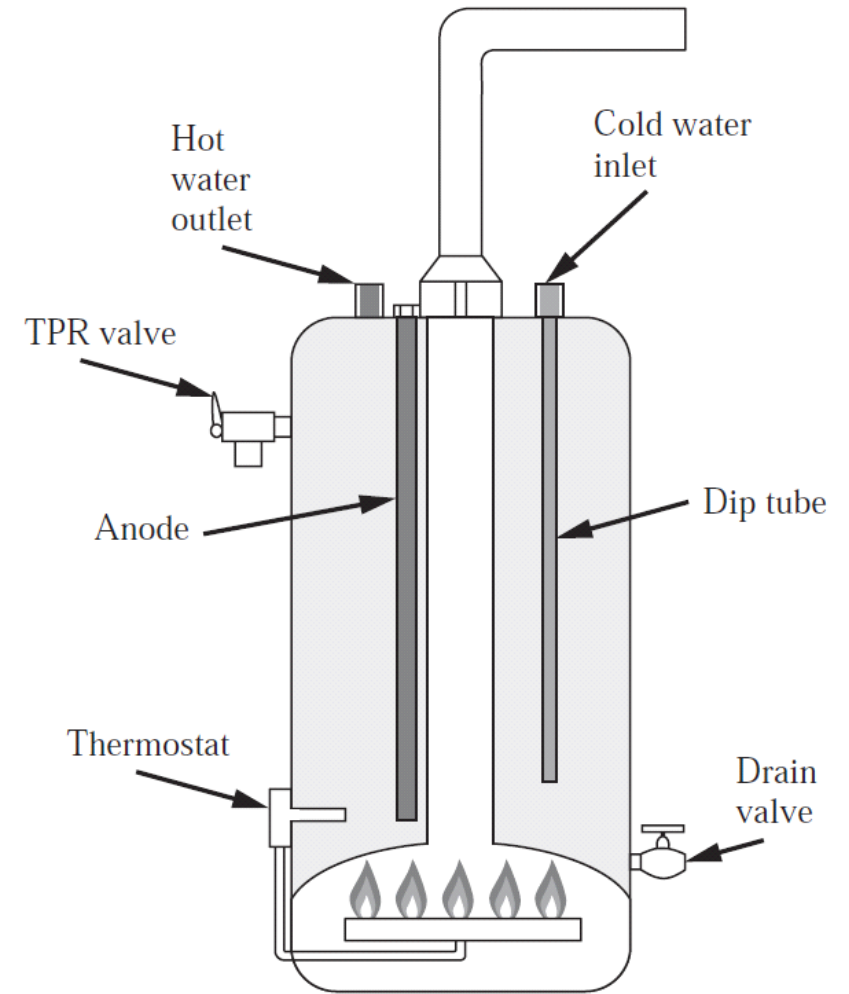


FIGURE 1-2
Physical drawing of a hot water heater.

Block Diagram → Relationship among variables
→ Flow of information

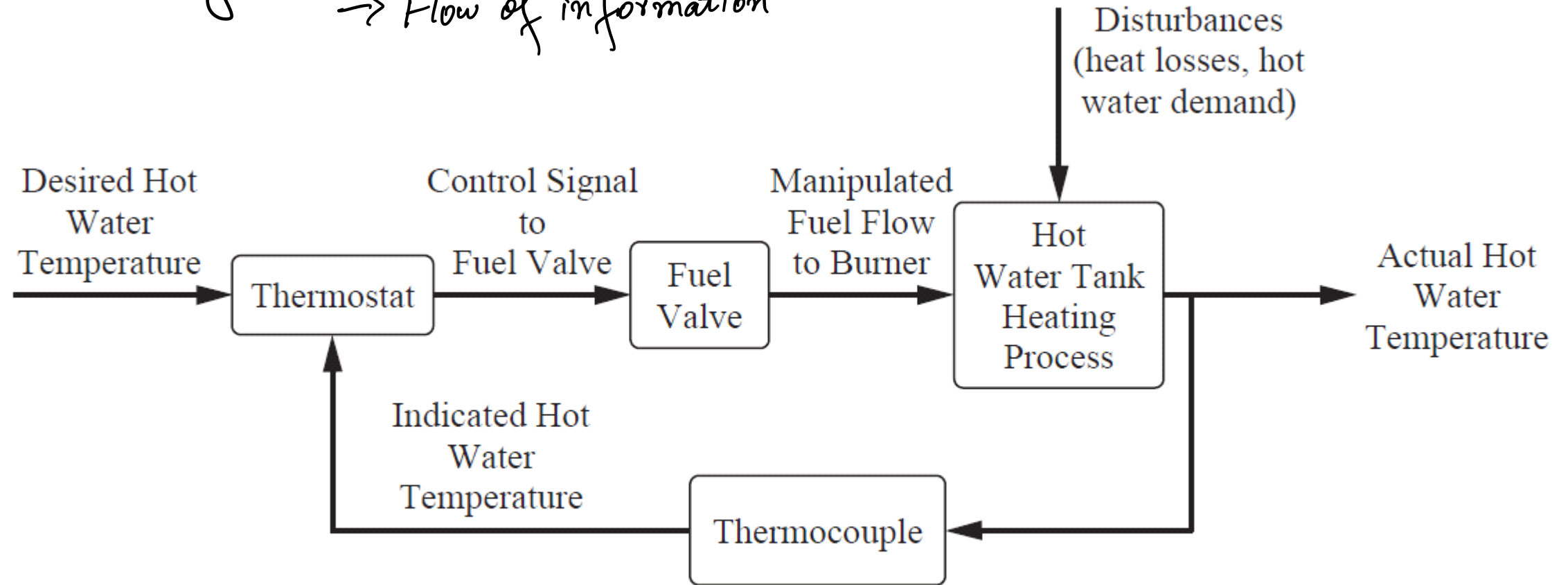


FIGURE 1-3

Block diagram of a hot water heater control system.

Types of Controllers

ON/OFF Type $\rightarrow$ ^{eg} Thermostat
 $\swarrow$ $\searrow$
 Full ON Full OFF

Controllers that can modulate signal based on

- $\rightarrow$ magnitude of error
- $\rightarrow$ how long the error persists
- $\rightarrow$ rate of change of error

Modulation of signal based on magnitude of error

⇒ Proportional change with error

→ Not able to attain zero error

⇒ some residual error persists

Modified by adding
Integral control

⇒ heat input changed by
additional amount proportional
to time integral of error

Further Problems

⇒ Measurement is not accurate and instantaneous

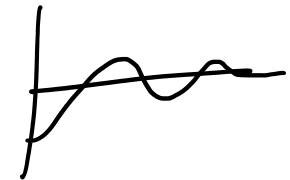
e.g. use of thermocouple

→ does not measure actual current fluid temp

(time required to stabilize)

⇒ Controller therefore acts on apparent error
instead of actual error

⇒ Superior Response than
Proportional control only
→ Can cause oscillations



Home work

1) Draw Block Diagram for an automobile cruise system.